

## Project Design Phase-II Technology Stack (Architecture & Stack)

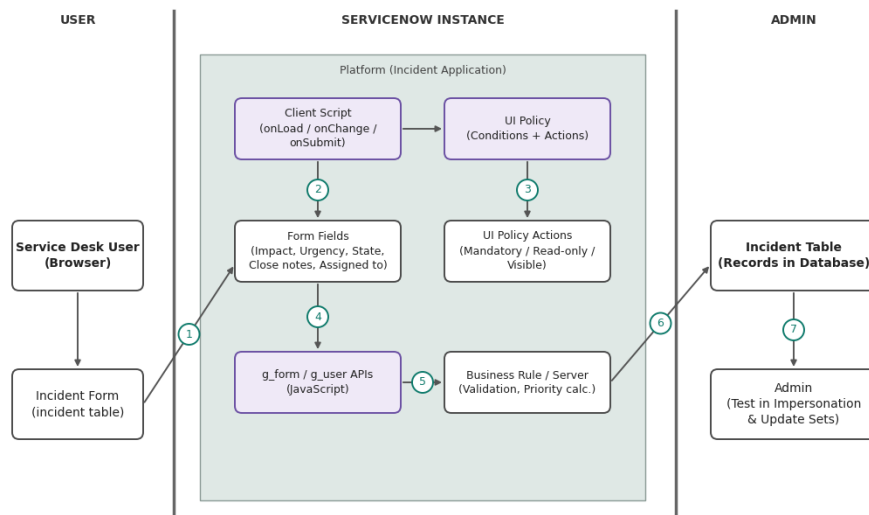
|               |                                                  |
|---------------|--------------------------------------------------|
| Date          | 30 September 2026                                |
| Team ID       | SWTID-2026-9596                                  |
| Project Name  | Implement Client Script and UI Policy (Incident) |
| Maximum Marks | 4 Marks                                          |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

### Example: Client Script and UI Policy on the Incident form (ServiceNow)

Reference: <https://developer.servicenow.com/>



### Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

**Table-1 : Components & Technologies:**

| S.No | Component                           | Description                                                                                                                                                                                              | Technology                                                      |
|------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1.   | User Interface                      | How the user works with the Incident form e.g. Service Desk / Native UI form, Service Portal, Mobile App.                                                                                                | ServiceNow Platform UI (Next Experience), HTML, CSS, JavaScript |
| 2.   | Client Script (Application Logic-1) | Runs in the browser to check or change field values (onLoad, onChange, onSubmit)                                                                                                                         | JavaScript (g_form, g_user APIs)                                |
| 3.   | UI Policy (Application Logic-2)     | Makes fields Mandatory, Read-only or Visible based on conditions, without coding                                                                                                                         | UI Policy (Conditions and Actions), UI Policy Script (optional) |
| 4.   | Business Rule (Application Logic-3) | Server-side check that works even if the form is bypassed (e.g. import, API)                                                                                                                             | Business Rule (JavaScript, GlideRecord)                         |
| 5.   | Database                            | Incident table (incident) with fields like Caller, Impact, Urgency, State, Close notes                                                                                                                   | ServiceNow Relational Database (Task table extension)           |
| 6.   | Cloud Database                      | Incident records are stored in the ServiceNow cloud instance                                                                                                                                             | ServiceNow Cloud Instance Database                              |
| 7.   | File Storage                        | Attachments added to incident records                                                                                                                                                                    | ServiceNow Attachment table (sys_attachment)                    |
| 8.   | External API-1                      | Not required for this activity (can send incident data to other tools if needed)                                                                                                                         | ServiceNow REST Table API (optional)                            |
| 9.   | External API-2                      | Not required for this activity                                                                                                                                                                           | Not applicable                                                  |
| 10.  | Machine Learning Model              | Not used in this activity                                                                                                                                                                                | Not applicable (Predictive Intelligence can be added later)     |
| 11.  | Infrastructure (Server / Cloud)     | Application runs on the ServiceNow cloud instance (Personal Developer Instance)<br>Local Server Configuration: Not required<br>Cloud Server Configuration : ServiceNow Personal Developer Instance (PDI) | ServiceNow Cloud (PDI), Update Sets                             |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                                          | Technology                                         |
|------|--------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1.   | Open-Source Frameworks   | Client Script, UI Policy and Business Rule are built-in platform features (no extra framework needed)                | ServiceNow Platform, JavaScript                    |
| 2.   | Security Implementations | Only allowed roles can create or edit scripts and incidents; Business Rule checks data again on the server           | ACLs, Roles (itil, admin), Secure HTTPS connection |
| 3.   | Scalable Architecture    | 3-tier architecture (browser, application server, database). Scripts run in the browser, so server load stays low    | ServiceNow Cloud Platform                          |
| 4.   | Availability             | Instance stays available through load balanced nodes and backup data centers                                         | Load Balancer, Data Center Failover                |
| 5.   | Performance              | Keep scripts short, use UI Policy instead of script where possible, and call the server (GlideAjax) only when needed | UI Policy, GlideAjax, Browser Caching              |

#### References:

<https://c4model.com/>

<https://developer.servicenow.com/>

<https://www.servicenow.com/docs/>

<https://www.servicenow.com/community/>

<https://www.servicenow.com/products/platform.html>